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Heparan sulfate proteoglycans in pathological protein aggregation and brain functionality

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Abstract

Proteoglycans are highly specialized proteins with specific interactive modules in their core proteins, allowing them to engage with a diverse range of cellular and structural proteins, contributing to many varied roles in cell processes. This review examines the roles of heparan sulfate (HS) proteoglycans (HSPGs) in the assembly of pathological protein aggregates in brain tissues, which negatively impact cognitive brain function. While HSPGs play roles in normal brain development, they also contribute to the abnormal accumulation and persistence of protein aggregates in Alzheimer's and Parkinson's diseases (AD and PD), prolonging their neurotoxic effects. HSPG-mediated effects on the innate immune system may also contribute to the development of neuroinflammation and pathogenesis of neurodegenerative changes in the central nervous system. These pathological developments lead to impaired cognition, permanent memory loss in dementia and AD, and an inability of the brain to exert normal neuromuscular control, affecting motor functions and body movement in PD. A deeper understanding of the properties of HS in these degenerative processes is essential for the development of therapeutic measures to treat these conditions. Significant improvements in neuroimaging, next-generation glycosaminoglycan analytical techniques, and functional glycomics hold great promise for elucidating the complexities of HS structure and function. These advancements may significantly aid in the development of HS biotherapeutics to treat these debilitating conditions, which are increasingly impacting the aging global population. While a number of promising therapeutic candidates have already emerged, further research is required to optimize their biotherapeutic applications in this challenging area of pathobiology, bioregulation, and the recovery of neural function.

Keywords: Amyloid; Heparan sulfate proteoglycans; Synuclein; Prion; Tau; Alzheimers disease; Parkinsons disease; Neurocognitive decline

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1. Introduction

Proteoglycans are highly specialized proteins encoded by a small fraction of the human genome, with approximately 0.25% of all protein-encoding genes responsible

for their production. There are around 20,000 proteoglycan-associated genes listed in major human genome databases (GENCODE/Ensembl, RefSeq, and UniProtKB).[1] Approximately 50 proteoglycans, including 18 heparan sulfate (HS) proteoglycans (HSPGs), have been identified, and many excellent reviews have explored their properties.[2-5] These reviews highlight the diverse roles of HSPGs in various biological processes, including cell differentiation and proliferation, tissue development and morphogenesis, angiogenesis, skeletogenesis, and extracellular matrix (ECM) remodeling during tissue repair. HSPGs also play critical roles in interactions with growth factors and morphogens during tissue development, matrix stabilization, and cell-matrix communication, all of which are crucial for regulating tissue homeostasis. HS is an ancient glycocalyx molecule[6] that evolved over a 500-million-year period as a highly diverse and essential cell regulatory molecule.[7] Natural selection has favored HS for its versatility in molecular recognition, information storage, and information transfer.[8] The interactive properties of HSPG stem from the glycocode of their glycosaminoglycan (GAG) side chains[9] and specific interactive modules within their core proteins, allowing them to interact with a wide range of cellular and structural proteins (the HS interactome). The aim of this review is to elucidate the role of HSPGs in the propagation and deposition of pathological protein aggregates in the brain, which contribute to the development of neurodegenerative processes and diseases associated with cognitive decline.

1.1. HS biosynthesis

During the biosynthesis of HS, D-glucuronic acid (GlcA) and *N*-acetylglucosamine (GlcNAc) are assembled in an alternating fashion to form the repeating GlcA-GlcNAc disaccharide, which constitutes the backbone of HS.[10] The nascent HS chain undergoes further remodeling through the actions of several families of modifying enzymes, adding complexity to the HS structure. One of the initial steps in this process is the removal of the acetyl group from GlcNAc, followed by the addition of sulfate groups. D-glucuronic acid can undergo epimerization to L-iduronic acid (IdoA), and its uronic acid groups can be *O*-sulfated at the C2 position. Glucosamine can also be *O*-sulfated at the C6 position and less commonly at the C3 position.[10] The spatial and temporal regulation of glycosyltransferases and other HS-biosynthetic enzymes within tissues accounts for the diversity in HS structure observed during tissue development and in pathological conditions. Sulfation along the HS chain is not uniform; HS chains contain regions of high sulfation interspersed with domains of high acetylation, non-sulfated regions, and less-modified de-acetylated regions. Heparin, a specialized form of HS, lacks these less-modified regions and is the most heavily modified GAG. HS biosynthesis is not a template-driven process; rather, it is guided by the activity of HS biosynthetic enzymes that are spatiotemporally expressed in tissues, contributing to the diversity of HS functional forms.[11] HS is assembled from at least 12 different disaccharides, which are variably distributed along an HS chain.[10] The fine structure of HS is remarkably diverse, theoretically allowing for an infinite number of protein-binding epitopes, with over 1,000,000 structural epitopes possible in an HS octasaccharide.[12,13] Cummings[12] has calculated that for a typical interactive HS pentasaccharide, based on repeat disaccharides containing GlcA, IdoA, and IdoA2S at position 1, and GlcNAc, GlcNS, GlcNAc6S, GlcNS6S, GlcNS3S, and GlcNS3S6S at position 2, there are 972 ($3 \times 6 \times 3 \times 6 \times 3$) possible structural combinations for one pentasaccharide and 1,944 ($6 \times 3 \times 6 \times 3 \times 6$) combinations for other, making a total of 2,916 possible pentasaccharide structural combinations. The spatiotemporal expression of defined HS structures and the modular core protein components of HSPGs are tightly regulated during embryonic development, skeletal maturation, and the specific features of pathological tissues.

2. HS sulfation as a key functional determinant of proteoglycans

A significantly higher 6*O* sulfation level in the HS chains of syndecan (SDC)-2 compared to those of SDC-4 correlates with increased reactivity of SDC-2 toward vascular endothelial cell growth factor (VEGF)-2, leading to enhanced activation and elevated neovascularization. This suggests that specific HS sulfation patterns contribute to the regulation of distinct biological activities *in situ*.[14] Despite the importance of GAG multivalency in proteoglycan function, only a few studies have examined its biological significance. Two studies have proposed that dimerization or clustering of SDCs in focal adhesion complexes, along with the associated increase in charge density, drives receptor activation and interactive processes with fibroblast growth factor (FGF)2 and VEGF2.[15,16] This clustering increases FGF2-mediated cell signaling, cell adhesion, and neovascularization. The diversity of GAG side chain sequences attached to proteoglycan core proteins at the cell surface contributes to their cell-instructive and ECM-interactive properties. These sequences modulate interactions with growth factors, chemokines, morphogens, proteases, and protease inhibitory proteins, acting as cell-responsive cues delivered by the ECM.[11] The Wnt signaling pathway plays an important role in regulating embryonic tissue development and adult tissue homeostasis. Wnt proteins, which are secreted lipid-modified glycoproteins, form morphogen gradients that direct cellular behavior.

However, due to their hydrophobic nature and poor solubility, Wnt proteins require HSPGs for proper activity. HSPGs maintain solubility and facilitate the transport of Wnt proteins, enabling the formation of gradients in tissues that drive developmental processes. Specifically, Wnt proteins bind to the low-density lipid receptor domain II of perlecan, which acts as a Wnt delivery vehicle during tissue development.

2.1. The functional role of the HS glycocode and the enigmatic 3-*O* HS sulfation motif

HS GAG chains are typically depicted as linear structures, but they are actually helical assemblies with unique small "kink(s)" where GlcA is epimerized to IdoA. The glycosidic linkages around IdoA offer greater chain flexibility, allowing for enhanced interaction of 3-*O* sulfation motifs with a diverse range of ligands.[15] Although 3-*O* sulfation in HS is rare, it plays a crucial role in ligand interaction, notably with antithrombin, Wnt, sonic hedgehog, SARS-CoV2 spike protein, FGF receptor (FGFR), tau, and neuropilin-1. HS displays the most diverse "sulfation codes" of any GAG, enabling selective recognition and interaction with numerous proteins,[12] as reflected in HS and GAG interactome databases. Specific HS sequences have been identified that facilitate interactions with Wnt, lipoprotein lipase, antithrombin, and FGF-FGFR. The sulfation of HS arises from the controlled spatiotemporal expression of a superfamily of N- or O-sulfotransferases. Seven isoforms of 3-O-sulfotransferases (HS3ST-1, -2, -3_A, -3_B, 4, -5, and -6) introduce 3-*O*-sulfation motifs into HS chains in the final biosynthetic step. Despite the relative scarcity of 3-*O* sulfation, its presence underscores its importance as an interactive motif.[16] Tau glycoprotein[17] and SARS-CoV-2 spike glycoprotein[18] have been shown to preferentially bind to the 3-O-sulfation motif in HS. The sulfation sequences flanking the 3-*O*-sulfation motif also contribute in an accessory capacity to the specificity of these interactive 3-*O* HS sequences, equipping HS with a diverse dynamic recognition system through its 3-*O*-, 2-*O*-, and 6-*O*-sulfation motifs. This enables precise identification of cell regulatory molecules and activation of growth factor receptors through a sulfation code, which is reflected in the co-receptor functions of cell surface HSPGs.[18]

Tau is one of the few proteins that preferentially bind to the 3-*O*-sulfated HS isoform.[19] The formation of αsynuclein (α-syn) and βamyloid (Aβ) fibrils each requires *O*-sulfation, as the removal of *N*-, 6-*O*-, or 2-*O*-sulfation does not significantly inhibit this aggregative process, highlighting the significant role of 3-*O* sulfation.[20] The binding of tau, Aβ, and αsyn to neuronal cell surface HSPGs promotes their cellular uptake and the seeding of protein aggregates in the brain. The GAG-binding requirements of recombinant tau, αsyn, and Aβ fibrils with HS have been determined using a small heparin mimetic library, which identified the critical size and sulfation positions required for the formation of such protein aggregates.[20] This research has clarified the HS size and sulfation patterns required to promote aggregate propagation. Specifically, tau aggregates require a precise HS architecture containing *N*-, 6-*O*-, and 3-*O*-sulfated residues. In contrast, the binding of α-syn and Aβ aggregates is less dependent on these specific sulfation patterns. The internalization of tau protein and the propagation of its aggregates are regulated by 6-*O*-sulfation patterns on cell surface HSPGs, while 3-*O* sulfation of HS also enhances tau interaction and cellular uptake.[15] Moreover, cell surface HSPGs play crucial roles in the cellular attachment, subsequent internalization, fibril propagation, and spreading of both tau and α-syn fibrils.[21-24]

3. Lipid metabolism and amyloid deposition in the Alzheimer's disease (AD) brain

The brain has the second-highest lipid content of all human tissues, following adipose tissues, and lipids account for 50% of the brain's dry weight. Myelin, which functions as axonal insulation to ensure ultra-high conduction velocity of neural signaling, has significantly higher lipid content (78 – 81% dry weight) compared to white matter (49 – 66%) or gray matter (36 – 40%). Accumulation of different metals in brain tissues can lead to the production of peroxides, reactive oxygen species, and free radicals, resulting in lipid peroxidation and mitochondrial dysfunction, which can impair normal brain function.[25] Minimizing lipid peroxidation through dietary flavonoids, which possess antioxidant and metal-chelating properties, inhibit amyloid fibril assembly,[26] and can be transported through the gut-brain axis, holds promise as a therapeutic approach for neurodegenerative diseases. Apolipoprotein E2 (*APOE2*) gene therapy has been shown to reduce amyloid deposition and neuroinflammation. The APOE isoformes—APOE2, APOE3, and APOE4—interact with Aβ1-40, promoting its cellular retention. Notably, individuals carrying the ε4 APOE allele are at a greater risk of developing AD and cerebral amyloid angiopathy, which are associated with age-related cognitive decline.

4. HS proteoglycans and diseases of cognitive decline

The brain contains a diverse array of proteoglycans (Table 1). HSPGs associated with pathological neural tissues are implicated in the deposition of plaques and fibrillary tangles involving tau protein, Aβ fibrils, and

Table 1. Impact of heparan sulfate proteoglycan (HSPG) and β-amyloid (Aβ) interactions on amyloid deposition and tissue properties in Alzheimer's disease (AD)

HSPG	Expression in AD	Interactions with Aβ	Effects on tissues
Perlecan	Elevated levels in AD[56]	Inhibits amyloid degradation *in vitro*[57] and promotes amyloid deposition	Perlecan domain V inhibits α2β1 and α5β1 integrin-mediated Aβ toxicity[58-61]
Agrin	Agrin is abundant in Aβ plaques and elevated in CSF of AD patients	Binding of agrin to Aβ increases and stabilizes fibril formation	Elevated endothelial agrin Aβ deposition regulates BBB permeability
GPC 1-6	GPC 1-6 are downregulated in the cerebellum but upregulated in limbic areas in AD.[62]	Aβ treatment of neurons and astrocytes amplifies GPC1 transcription and sulphation; GPC4 aids Aβ internalization and toxicity	Alterations in the distribution of GPC4 lead to cognitive and social disorders[63]
SDC1 – 4	Hippocampus SDC3 and SDC4 are elevated in AD, corresponding with disease severity	Neuronal SDC3 plays a critical role in Aβ1 – 42 endocytosis[64]	SDC3 contributes to long-term potentiation in the hippocampus[65]
Collagen XVIII (COL18A1)	Upregulated expression in the entorhinal cortex and hippocampus in severe AD[40]	Endostatin forms cytotoxic amyloid fibrils *in vitro*;[66] Coll XVIII forms vascular amyloid deposits and senile plaques in AD.[67]	Dysfunctional ECM in AD is linked to COL18A1 mutations, which lead to Knobloch syndrome and brain abnormalities

Abbreviations: BBB: Blood–brain barrier; CSF: Cerebrospinal fluid; ECM: Extracellular matrix; GPC: Glypican; LRP-1: Low-density lipoprotein receptor-related protein-1; SDC: Syndecan.

abnormally folded α-syn, all of which detrimentally impact neuronal viability and function.[19] Parkinson's disease (PD), the second most common neurodegenerative disorder after AD, is characterized by a loss of functional brain dopaminergic neurons in the brain, leading to impaired motor control. Aggregated forms of α-syn are abundant in pathological PD tissues and are involved in its pathogenesis.[27] Numerous studies have suggested that HSPG interactions with tau contribute to various stages of the development of AD, including seeding, aggregation, plaque formation, neurofibrillary tangle-mediated toxicity, and delayed clearance of amyloid deposits.[28] The oligomeric Aβ fragment activates hyperphosphorylation of the tau protein. In a healthy brain, tau maintains axonal stability, ensuring efficient transmission of action potentials during neurotransduction. However, hyperphosphorylation of tau disrupts axonal transmission, leading to cytotoxic signaling. Recent research has proposed that HSPG-tau assemblies contribute to the formation of tunneling nanotubes in brain tissues, facilitating the intercellular transport of tau and Aβ peptides. These nanotubes play a role in the seeding of amyloid deposits and the pathogenesis of AD.[29]

5. Central nervous system (CNS) proteins with Kunitz protease inhibitor (KPI) modules in the delayed clearance of pathological protein aggregates in AD and PD

CNS proteoglycans containing core proteins KPI modules possess tissue-protective properties. However, when these proteoglycans are associated with pathological protein aggregates, they may impede the clearance of deposits observed in AD and PD. HSPGs such as agrin and perlecan have been shown to associate with Aβ and α-syn aggregates, thereby promoting their propagation and persistence in AD and PD.[30]

5.1. Inter-*α*-trypsin inhibitor (ITI)/bikunin and related Kunitz domain proteins in brain tissues

ITI is localized in the nuclei of neurons and astrocytes across all age groups, with cytoplasmic expression becoming more prominent in adulthood.[31] Members of the ITI family are components of the innate immune system within the CNS[31] and exhibit neuroprotective properties.[32] In neonatal rats, ITI levels decrease following hypoxic-ischemic brain injury. Bikunin, the light chain of ITI, contains two KPI domains that inhibit pancreatic trypsin, chymotrypsin, leukocyte elastase, and cathepsin G, kallikrein, and plasmin. Bikunin has been identified in the hippocampus and cerebral cortex and has also been detected in human brain tumors.[33] Other related Kunitz domain proteins include tissue factor pathway inhibitors α, β, and δ, as well as the multidomain transmembrane serine protease inhibitors HAI-1 and HAI-2 (hepatocyte growth factor [HGF] activator inhibitors 1 and 2). These multidomain proteins are neuroprotective and membrane-bound and play roles in maintaining the integrity of the neural basement membrane and brain microvasculature, in addition to exhibiting anticoagulant activity.

5.2. Tissue factor protease inhibitors (TFPI) 1 and 2 in the brain

TFPIα is a protein weighing approximately 43 kDa and contains three tandem KPI domains. The TFPIβ isoform

(32kDa) lacks the KPI-3 domain. TFPI functions as an anticoagulant protein that blocks the initiation of blood coagulation by inhibiting the tissue factor/activated factor VIII (TF–fVIIa) complex and thrombin activation. A study has demonstrated that Tfpi$^{-/-}$ mice die during embryonic development, displaying stunted brain growth, delayed meningeal development, and severe vascular pathology.[34] The generation of thrombin is associated with hemorrhagic stroke and is central to the pathogenesis of acute ischemic stroke. TFPI-2 exhibits neuroprotective properties, inhibits human glioblastoma formation, and inhibits a range of proteases, including plasmin, trypsin, chymotrypsin, cathepsin G, plasma kallikrein, FVIIa, Fxa. In addition, TFPI-2 inhibits fibrinolysis, thereby regulating pericellular proteases that are implicated in cancer cell invasion, vascular inflammation, and atherosclerosis. *TFPI2* has been proposed as a tumor repressor gene in several cancers, including glioma (Figure 1).

5.3. HAI-1 and HAI-2 in the brain

HGF is a potent mitogen with multifunctional effects on various cell types. The activity of HGF *in vivo* is regulated by HGF activator protease, which is in turn regulated by hepatocyte activator serine protease inhibitors HAI-1 and HAI-2.[35] Astrocytes in the white matter produce these proteins. HGF/c-MET (mesenchymal-epithelial transition) signaling promotes cell proliferation, survival, migration, and motility of cancer cells, as well as neocortical neuronal growth and synaptogenesis. HAI-2 specifically inactivates the HGF activator, thereby inhibiting the conversion of pro-HGF into its active form.

5.4. Amyloid precursor protein (APP)

The APP contains a KPI domain (protease nexin 2), which acts as a highly specific substrate for mesotrypsin, atypical human trypsin implicated in a number of neurodegenerative disorders[36] (Figure 2A). Neurotrypsin also cleaves agrin locally at the synapse. The KPI domain in APP inhibits pro-neuropeptide processing, and the accumulation of APP in Aβ plaques may hinder their clearance from tissues, thereby prolonging their neurotoxic effects in AD and PD. The accumulation of Aβ arises from the processing of APP-1 by β- and γ-secretases in the brain, an early toxic event in the pathogenesis of AD that leads to the formation of disruptive intracellular neurofibrillary tangles and extracellular Aβ senile plaques. APP is also cleaved by α-secretase, forming a soluble APP alpha (sAPPα) peptide through a non-amyloidogenic pathway that improves neuronal survival, attenuates neuronal apoptosis, and has neuroprotective effects.[37] Alpha secretases are members of the ADAM (a disintegrin and metalloprotease) family, which are expressed on cell surfaces and anchored to the cell membrane. Physiological levels of sAPPα exhibit neurotrophic effects and play critical roles in neurite outgrowth and brain functional development through the activation of long-term potentiation (LTP). LTP is a process where synaptic signaling is progressively strengthened over time, leading to increased synaptic

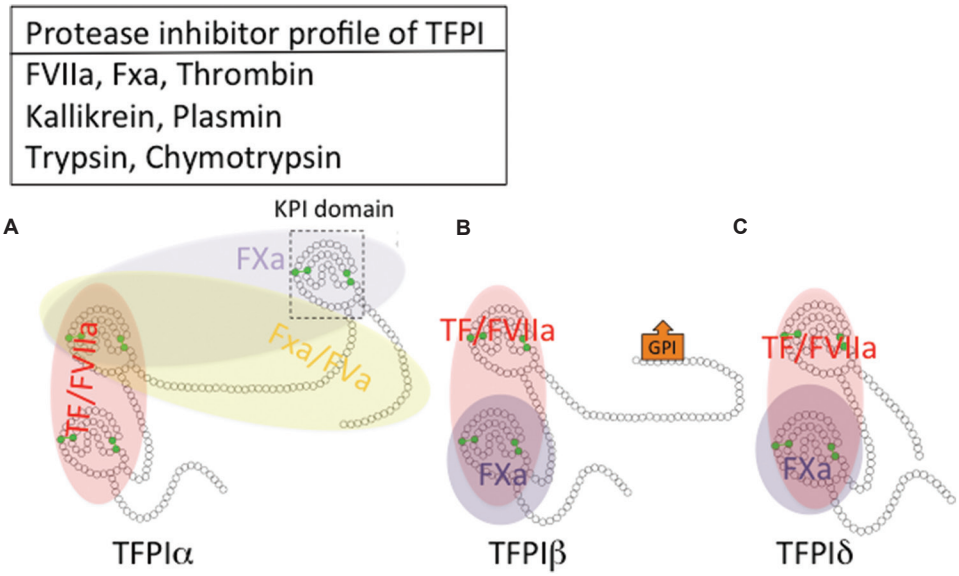


Figure 1. Schematic depiction of the structural organization of tissue factor protease inhibitors (TFPI) α, β, and δ, showing their Kunitz protease inhibitor domains (KPIs), which are colored-coded (A-C). The proteases they inhibit are listed (D). TFPIβ (B) is anchored to the cell surface through a glycosylphosphatidylinositol anchor. The molecular domains that inhibit tissue factor/factor VIIa (TF/FVIIa), factor Xa/factor Va (FXa/FVa), and factor Xa (FXa) are indicated in the shaded regions.

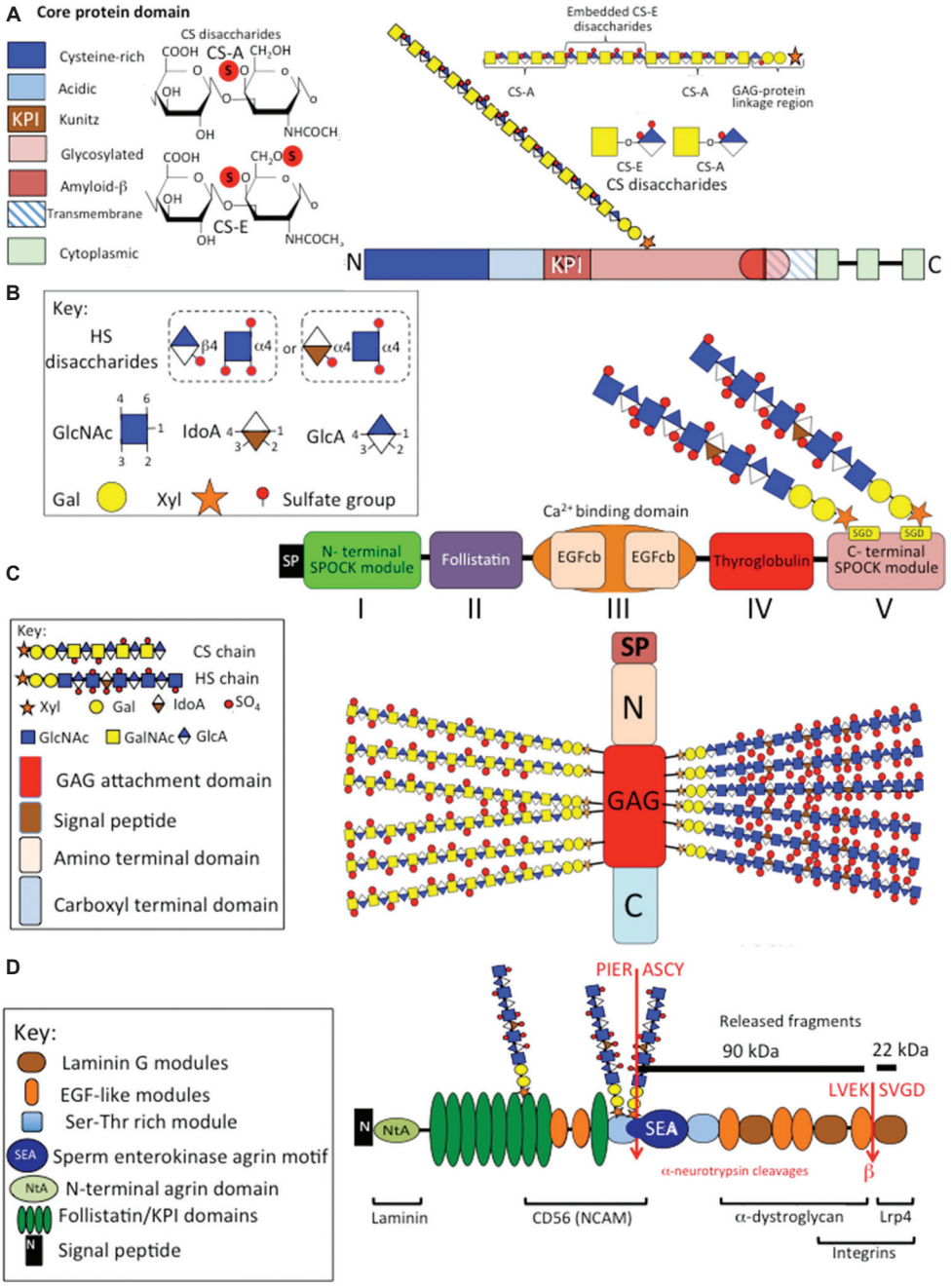


Figure 2. Schematic depictions of the structural organization of appican (A), testican (B), serglycin (C), and agrin (D), proteoglycans that play documented roles in the deposition of insoluble protein aggregates in neurodegenerative diseases such as AD and PD. For appican (A), the diagram illustrates the sulfation of its chondroitin sulfate chain, highlighting the embedded chondroitin sulfate E disaccharides within the chondroitin sulfate A chain, and the sulfation positions of the chondroitin sulfate A and E disaccharides. The core protein domain organization of appican is also depicted, showing its extracellular cysteine-rich N-terminal domain, a central Kunitz protease inhibitor (KPI) domain, a β-amyloid peptide module that self-assembles to form amyloid aggregates, a transmembrane domain, and a C-terminal cytoplasmic domain. The domain structure of the testican core protein (B) is presented, featuring its N-terminal SPOCK (secreted protein acidic and rich in cysteine ([SPARC])/osteonectin and kazal-like domain proteoglycan) domain, a cysteine-rich module homologous to follistatin, central Ca^{2+}-binding epidermal growth factor (EGF)-like domains, a thyroglobulin domain, and a C-terminal SPOCK module with two attached HS chains. The diagram also indicates the sulfation positions of the HS disaccharides and iduronic acid (IdoA)-substituted disaccharides within the HS chains of testican. For serglycin (C), the heavily substituted central glycosaminoglycan (GAG) domain is shown, which can contain HS or oversulfated chondroitin sulfate (CS) chains, along with other core protein domains. Finally, the structural organization of the HS proteoglycans, agrin (D), is illustrated, showing its eight follistatin modules, multiple EGF-like domains, a central sperm-enterokinase-agrin)SEA) domain, and a Ser/Thr-rich module with HS substitution sites and laminin G interactive sites. The interactive sites in the agrin core protein for laminin, neural cell adhesion molecule (NCAM, CD56), α-dystroglycan, low-density lipoprotein receptor-related protein (LRP) 4, and integrins are also depicted. In addition, the two cleavage sites in the agrin core protein targeted by α-neurotrypsin are indicated.

stability. The accumulation of Aβ generated from APP through an amyloidogenic pathway involving consecutive cleavages by β-secretase and Q-secretase is causally linked to the pathogenesis of AD.[38] The co-deposition of HS with Aβ in plaques in the AD brain accelerates pathological neurodegenerative changes in brain tissues.

5.5. Testicans regulating matrix metalloproteinases

Testican-1, -2, and -3, also known as SPOCK (secreted protein acidic and rich in cysteine [SPARC]/osteonectin and kazal-like domain proteoglycan), are members of a widely distributed modular five-domain proteoglycan family that contains two C-terminal HS chains, extensively expressed in brain tissues (Figure 2B). Domain I of testicans is a unique N-terminal domain, while domain II is a cysteine-rich module homologous to follistatin/kazal-type protease inhibitors. Testican-1 and -2 are expressed by multiple neuronal cell types in the olfactory bulb, cerebral cortex, thalamus, hippocampus, cerebellum, and medulla.[39] Testican-1 modulates matrix metalloproteinase (MMP) activation by inhibiting membrane-type MMPs and cathepsin L. In contrast, testican-2 abrogates the inhibition of membrane type 1-MMP- or membrane type 3-MMP-mediated pro-MMP-2 activation by testican-1. Testican-3, an HSPG exclusive to brain tissues, suppresses membrane type 1-MMP-mediated activation of MMP-2. Increased levels of testican-1 have been observed in AD brain tissues, where it accumulates in co-aggregates with Aβ plaques in the frontal, temporal, and entorhinal cortices of AD patients.[40] This accumulation impedes the clearance of Aβ from AD brains, thereby impacting normal brain function.

5.6. Bikunin

Bikunin contains two KPI domains that may inhibit the clearance of Aβ deposits. These KPI domains inhibit a range of serine proteases, including trypsin, chymotrypsin, leukocyte elastase, cathepsin G, kallikrein, and plasmin. The inhibition of these serine proteases prevents the activation of MMPs.[41]

5.7. Serglycin

Serglycin (SRGN) is widely distributed in brain tissues, as shown in the Human Protein Atlas (http://www.proteinatlas.org/ENSG00000122862-SRGN/brain). SRGN is an intracellular secretory granule proteoglycan found in neurons, endothelial cells, endocrine cells, and hematopoietic cells. It carries 7 – 15 GAG chains, each over 100 disaccharides in length, attached to a central core protein region composed of 9 – 24 Ser-Gly repeats, giving the protein its name (Figure 2C). These Ser-Gly repeats are exceptionally resistant to proteolytic degradation, and along with their attached GAG chains, they package active proteases in intracellular granules in an inactive form. *SRGN* colocalizes with plasminogen activator in secretory vesicles, regulating plasminogen activator secretion and MMP activation. Microglial SRGN is upregulated in the ischemic brain.[42] SRGN's interactions with properdin promote the production of proinflammatory mediators,[43] thereby activating the complement system and innate immune response.[44] Properdin, secreted by neutrophils, monocytes, and T-cell thymocytes, accumulates in the ischemic brain and acts as a damage-associated molecular pattern that exacerbates neuroinflammation in the injured brain. T-cell infiltration through the blood–brain barrier (BBB) promotes AD and cognitive decline. The innate immune response thus contributes to CNS neurodegeneration. In AD, the BBB may be impaired, leading to elevated T-cell and mast-cell infiltration.[45]

5.8. Agrin

Agrin contains nine follistatin KPI domains, which may confer resistance to the proteolytic clearance of protein aggregates in AD, thereby prolonging the duration of neurotoxic effects.[46] Agrin also has a laminin-binding domain that is structurally related to tissue inhibitor of MMP-1,[47] and it features multiple epidermal growth factors (EGF)-like and laminin G modules, a central sperm-enterokinase-agrin (SEA) domain, and a Ser/Thr-rich module that contains the HS substitution sites (Figure 2D). Agrin interacts with laminin, neural cell adhesion molecule (NCAM, CD56), α-dystroglycan, low-density lipoprotein receptor-related protein (LRP) 4, and integrins. α-neurotrypsin cleaves agrin at two cleavage sites in the agrin core protein: PIER↓ASCY (a-cleavage site) and LVEK↓SVGD (b-cleavage site).

6. Propagation of pathological brain protein aggregates by HS proteoglycans

HS proteoglycans play essential roles in neurodegeneration associated with neuritic plaques and cerebrovascular blood vessels in AD.[30] SDCs, a type of HSPG, promote the cellular uptake and fibrillation of αsyn, tau protein, and Aβ 1 – 42, which contribute to the formation of senile plaques in AD. HSPGs act as mediators between monomeric tau and the activation of the ERK1/2 cell signaling pathway, facilitating tau internalization.[48] The aggregation of tau fragments and the formation of filaments is influenced by the sulfation code of HS in HSPGs,[19] particularly regulated by the 6*O*-sulfate content and promoted by 3*O*-sulfation.[49] In AD, HSPGs promote amyloid pathology by promoting the formation of amyloid aggregates and filaments, similar to their roles in the assembly of prion aggregates in Creutzfeldt-Jakob disease and scrapie. Amyloid fibrils, which are formed by several normally soluble proteins,

become resistant to enzymatic degradation when they interact with HS, resulting in the assembly of macro-aggregates that persist in brain tissues. To date, more than 36 plasma proteins have been identified that can form amyloid deposits. Despite the absence of a common amino acid sequence among these proteins, they all share the ability to self-assemble into stable, stacked βpleated sheet fibrillar structures. The HSPG binding site in amyloid is a 13- to 16-amino-acid segment containing the His-His-Gln-Lys sequence. Low-molecular-weight heparins have been shown to reverse this binding step, thereby inhibiting fibril formation, and have been suggested as potential therapeutic agents for treating amyloid plaque formation in AD.

6.1. Prion disease

Transmissible spongiform encephalopathies, or prion diseases, are a group of devastating neurological disorders characterized by neuronal loss, mental deterioration, and spongiform degeneration. Although rare in humans, prion diseases have a high prevalence in deer and elk and also occur in sheep and goats, where the condition is known as scrapie. Human prion diseases include Kuru, as well as sporadic, iatrogenic, and familial forms of Creutzfeldt-Jakob disease, Gerstmann-Sträussler-Scheinker disease, and fatal familial insomnia.[50] Kuru is a form of prion disease restricted to Papua New Guinea, spread by ritualistic familial cannibalism. The causative agent of prion disease is a misfolded form of prion protein, PrP^{Sc} (Figure 3A). The normal prion glycoprotein, PrP^{C}, is a glycosylphosphatidylinositol (GPI)-anchored cell surface protein widely expressed in the immune and nervous systems, and it serves as a key neuronal receptor for Aβ oligomers, mediating neurotoxicity, neurodegeneration, and the pathogenesis of AD. Misfolding of PrP^{C} is also associated with transmissible spongiform encephalopathies. A disintegrin and metalloproteinase (ADAM10) proteolytically releases PrP^{C} from the cell surface as a soluble form devoid of its GPI anchor, which is a potential biomarker of neurodegeneration and a modulator of prion disease. The shedding of PrP^{C} from human neurons by ADAM10 prevents the binding and cytotoxicity of Aβ oligomers. Confocal immunofluorescence microscopy has shown that normal PrP^{C} and gylpican (GPC)1 colocalized inside an immortalized neuronal cell line (GT11). This colocalization does not occur in scrapie-infected GT11 cells (ScGT11), where GPC-1 remains at the cell surface, separate from PrP^{Sc}. Scrapie infection stimulated GPC-1 autoprocessing, and silencing of GPC1 elevates levels of intracellular PrP^{Sc} aggregates in infected cells. GPC1 autoprocessing clears PrP^{Sc} from infected cells, promoting the deposition of prions in brain tissues. $Ext1^{+/-}$ mice with shortened HS chains contain a rare 3*O*sulfate motif,[17] which binds tau. Misfolding of cellular prion protein, PrP^{C}, into its amyloidogenic isoform, PrP^{Sc}, leads to the formation of infectious protein aggregates in prion diseases. Influenza A virus has been shown to cause the conversion of PrP^{C} into misfolded PrP^{Sc} infectious prion particles in mouse neuroblastoma cells.[51] Furthermore, the SARS-CoV-2 spike protein contains prion-like peptide sequences that are capable of propagating insoluble protein aggregates.[52]

6.2. Tau

Hyperphosphorylated tau protein, a microtubule-associated protein, is a major component of the insoluble helical and straight filaments found in AD and other neurodegenerative disorders. The hyperphosphorylation of tau is essential for filament formation and is a defining feature of AD. Highly sulfated GAGs present in nerve cells promote interactions with tau, leading to the formation of neurofibrillary tangles—interactions that are central to neuronal pathology in AD (Figure 3B).

6.3. Synuclein

Lewy bodies, which contain GAGs and αsyn protein aggregates, are key features of PD. HS is critical for the aggregation and cellular uptake of αsyn. Studies using high-speed atomic force microscopy, dynamic light scattering, and nuclear magnetic resonance have shown that the interaction between human prion protein and αsyn results in the formation of a disordered complex that suppresses further aggregation, thereby inhibiting amyloidogenesis.[53] HSPG interactions with tau facilitate the prion-like propagation of tau aggregates in AD, leading to insoluble neurofibrillary tangles and senile plaque formation in dementia, which are detrimental to normal neural function and disrupt brain cognitive processes.[19] The transcellular spreading of tau in AD is mediated by its binding to neuronal cell surface HS, which contains a rare 3*O*sulfate motif.[17] The addition of 3*O*sulfate HS 12mers to cell cultures significantly reduces the cell surface binding and internalization of tau (Figure 3C).

6.4. Amyloid

It has been more than 30 years since the hypothesis was proposed that neurodegeneration in AD is caused by the deposition of Aβ in plaques within brain tissue. The accumulation of Aβ in the brain has long been considered the primary driver of AD pathogenesis.[54] The formation of neurofibrillary tangles containing tau protein is thought to result from an imbalance between Aβ production and Aβ clearance. However, the repeated failure of clinical trials aimed at improving Aβ clearance from pathological tissues has resulted in a shift in focus for neurodegenerative

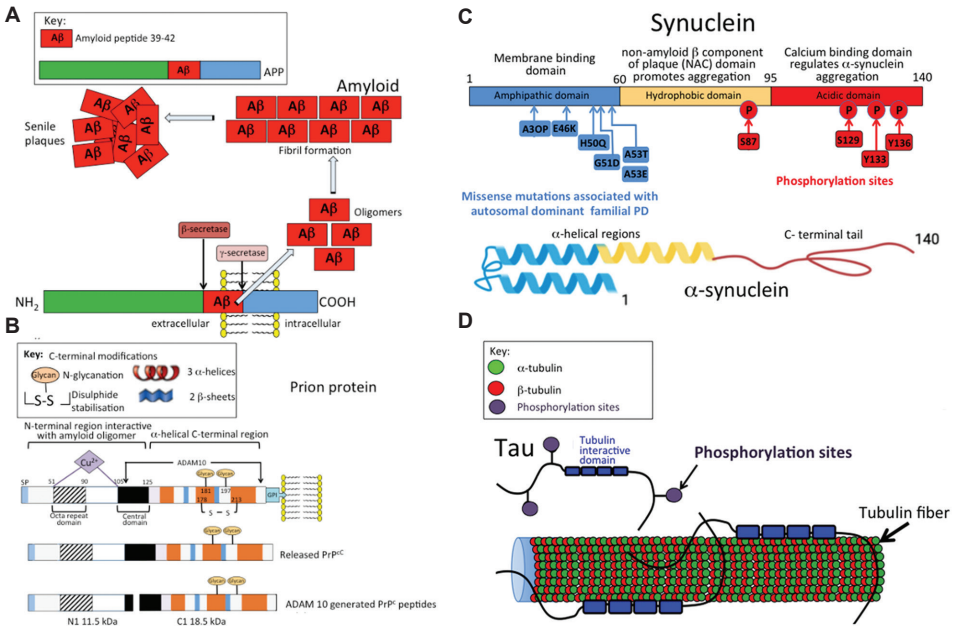


Figure 3. Structural organization of proteins involved in neurofibrillary tangles and extraneural plaque formations in Alzheimer's disease (AD) and Parkinson's disease (PD). (A) Enzymatic generation of amyloid βpeptide by β- and γsecretases and the subsequent extraneural oligomerization, fibril formation, and βamyloid plaque formation in AD. (B) Prion protein and its modular organization, identifying its functional regions and release from the cell surface, and the generation of prion N1- and C1-terminal peptides by ADAM10. (C) αsynuclein, depicting its functional domains that convey neurotoxicity and protein aggregate formation in PD. (D) Tau protein and its central tubulin-interactive modules that contribute to cytoskeletal stabilization and cell shape through the stabilization of microtubules.

therapeutics. Current research is increasingly focused on the role of highly sulfated HS, which promotes protein aggregate formation in amyloidosis. Interestingly, p53, a phosphoprotein tumor suppressor, also induces amyloid-like fibrils and aggregates in a prion-like manner.[55] Enzymatic remodeling of HS S-domains by sulfate-2 has been shown to downregulate the cellular uptake of p53 aggregates, offering a potential therapeutic avenue (Figure 3D).

7. Roles for specific HS proteoglycans in normal and neurodegenerative disease processes

7.1. CD47 and betaglycan HS/chondroitin sulfate hybrid co-receptors

CD47 is ubiquitously expressed in the CNS and regulates cell adhesion, proliferation, apoptosis, migration, tissue homeostasis, and the immune system. It functions as a temporally dually-modified HS/CS thrombospondin-1 co-receptor, modulating glioma cell invasion and angiogenesis.[68] Examination of synapse components in the normal and AD brain demonstrates elevated hippocampal pathologic tau and CD47 expression. Betaglycan (TGFBR3), another co-receptor, has a widespread distribution and is an important TGFβ receptor in the regulation of normal brain development.

7.2. Neurexins and neurodegeneration

Neurexins, which are HSPGs, stabilize synapses through interactions with cell adhesion synaptic glycoproteins, playing a critical role in neuronal function and signal transduction. There is a clear link between neurexins and neuropsychiatric disorders, including autism, schizophrenia, and neurodegenerative diseases. Changes in neurexin expression are observed in AD and PD.[69]

7.3. Syndecans and AD

Elevated expression of SDCs in AD is associated with the aggregation and modulation of cellular uptake of Aβ. APOE isoforms exert effects on AD development. Neuronal SDC3 increases the cellular uptake of APOEs; APOE2 enhances the internalization of Aβ, while APOE4 increases the extraneural buildup of Aβ plaques.[70] Intraneuronal accumulation of Aβ1 – 42 is an early feature of AD, with cell surface HSPGs promoting cellular uptake of Aβ1 – 42. SDC3 and SDC4 colocalize with amyloid plaques in AD, and the neuron-specific SDC3 isoform enhances the cellular uptake of Aβ1 – 42.

7.4. Glypicans and AD

Tau protein aggregates are a major driver of neurodegeneration and behavioral impairment. *APOE4*, the highest genetic risk factor for late-onset AD,

exacerbates tau hyperphosphorylation in mouse models. APOE4-mediated surface trafficking of APOE receptor, LRP-1, through GPC4 promotes tau spreading.[71]

7.5. Collagen XVIII

Collagen XVIII is well known for its role in stabilizing networks within basement membranes and the BBB. Mutations in collagen XVIII are associated with structural brain abnormalities in Knobloch syndrome.[72] Endostatin, a fragment of collagen XVIII, is upregulated in traumatic lesions in the brain and spinal cord. Structurally altered basement membranes and hydrocephalus have been noted in collagen XVIII-deficient mice. Aberrant neuronal deposition of endostatin occurs in vascular senile plaques and vascular amyloid deposits in AD.

7.6. Agrin

Agrin, which is associated with αsyn in Lewy bodies in PD, accelerates the formation of insoluble protofibrils. Agrin abnormalities in the microvasculature and brain parenchyma are linked to Aβ deposition in AD, contributing to the pathogenesis of both AD and PD through pathological lesions.[73]

7.7. Perlecan

Perlecan is immunolocalized within Aβ deposits in blood vessels in the AD brain and accumulates in fibrillar Aβ deposits, where it accelerates and stabilizes amyloid fibril formations. Domain V of perlecan interacts with the α2 integrin; Aβ is also a ligand for α2β1 and αvβ1, mediating Aβ-induced activation of cJun and caspase3 in the neurotoxic pathway in primary cortical and hippocampal neurons. Perlecan domain V modulates α2β1-mediated neurotoxic effects[58] and also interacts with the α5β1 integrin receptor and VEGFR2, increasing VEGF production and promoting angiogenesis, which improves BBB repair following stroke.[59] The HS chains of perlecan play critical roles in amyloid fibril assembly, both *in vitro* in microglia and *in vivo* in microglia/macrophages following the infusion of Aβ protein into the rodent hippocampus.[74]

8. Pathological features and treatment of insoluble protein deposits in brain tissues

8.1. Prion

Normal prion glycoprotein is a GPI-anchored protein located in the outer membrane of neurons. It has a flexible tail and a structured core that interacts with a wide range of partners, playing physiological roles in regulating the homeostasis of peripheral nerve myelination. Misfolding of PrP^C causes prion disease and mediates βoligomer-induced neurotoxicity in AD and PD. SDC1 mediates prion-like αsyn toxicity. The conserved central region of PrP^C acts as a linker between the N- and globular C-terminal domains, facilitating a regulatory metal ion-dependent interaction between the C- and Ntermini, which attenuates Nterminal toxicity.[75]

8.2. Tau protein

Tau protein has a molecular weight of 55 – 62 kDa and contains a central domain containing several tubulin-interactive modules, a Cterminal tail, and an Nterminal domain. Tau stabilizes microtubules in the cytoskeleton, which control cell shape. Tau contains multiple tubulin-interactive domains and up to 79 Ser and Thr phosphorylation sites. Pathological tau aggregation occurs in a large group of neurodegenerative diseases known as tauopathies. The formation of tau aggregates is triggered by two hexapeptide motifs within tau: plant homeodomain finger protein 6 (PHF6) and PHF6*. These motifs play critical roles in the development and preservation of the CNS. Tau can become dysfunctional through misfolding, and its accumulation in the brain can lead to AD, Down's syndrome, and dementia. The tau microtubular domain forms a core structure in the helical PHF6* and PHF6 filaments, promoting tau aggregation. Post-translational acetylation in PHF6* and PHF6 further promotes tau misfolding.[76] Direct inhibition of tau aggregation is a promising treatment strategy for the reduction of neurofibrillary tangles in AD. Tau protein inhibitory compounds, such as phenothiazines, have successfully modulated the pathophysiology of several neurodegenerative diseases.

8.3. Synuclein

α-synuclein is a small, unstructured, presynaptic protein that localizes in the nucleus and mitochondria of axonal terminal pre-synaptic neurons, where it interacts with tubulin and phospholipids and regulates neurotransmitter trafficking. In PD and dementia, αsyn aggregates into large protein deposits known as Lewy bodies, which are linked to neurotoxic pathways, including lysosomal, mitochondrial, and synaptic dysfunction, as well as endoplasmic reticulum and oxidative stress.[77,78] Tau deposits often occur alongside α-syn aggregates in PD brains. Tau accelerates the formation of α-syn fibril, contributing to the pathogenesis of PD. Tau-modified αsyn fibrils exhibit enhanced seeding properties, induce mitochondrial dysfunction, impair synaptic function, and contribute to neurotoxicity. Injection of tau-modified α-syn fibrils into animal models induces severe pathological changes in the brain, leading to motor dysfunction and cognitive impairment.[79] Tau knock-out models show attenuation of tau-mediated pathological changes in PD mice. Aggregates of αsyn

disrupt neuronal electrophysiological properties due to impairment in the Ca^{2+}binding properties of the α-syn N-terminal. Proper Ca^{2+} mobilization through depolarized neuronal membranes is fundamental to neuron activation and neurotransduction, which regulates the transport of synaptic vesicles containing neurotransmitters. Phenolic compounds have been shown to inhibit the aggregation of αsyn and reduce cellular and synaptic toxicities.[80] Flavonoids, known for their anti-inflammatory and antioxidant properties, are efficacious in the treatment of neurodegenerative processes[81] and can penetrate the BBB.[82] Analog flavonoid derivatives with improved efficacy target multiple neurodegenerative signaling pathways, including those involving nitric oxide synthase, cyclooxygenase-2, monoamine oxidase, and peroxide-induced neuronal death.

8.4. Amyloid

The APP is a highly expressed neuronal single-pass transmembrane proteoglycan that is processed by several proteases to release the Aβ peptide (consisting of amino acids 36 – 43), which forms Aβ plaques. Hyperphosphorylated forms of microtubule-associated tau protein are also prominent in these plaque formations. Misfolded Aβ can act as a template, promoting the misfolding of other β peptides, thereby seeding the production of Aβ protein aggregates and inducing similar changes in tau protein associated with aggregate Aβ1 – 40 and Aβ1 – 42 peptides. Soluble Aβ oligomers exhibit both extracellular and intracellular neurotoxic effects through interactions with membrane receptors and the formation of ion-permeable pores in membranes. The accumulation of Aβ peptides results in synaptic dysfunction and disturbances in intracellular Ca^{2+} homeostasis. Ca^{2+} is a key messenger in neuron activation and electrophysiological signaling. Disturbance of astrocyte Ca^{2+} signaling by Aβ can disrupt gliotransmitter release processes vital to astrocyte-neuron communication and homeostasis.[83]

Flavonoids significantly improve cognitive functions and inhibit or delay Aβ aggregation and neurofibrillary tangle formation in AD. Diets rich in flavonoids are associated with increased intellectual abilities and may postpone or hinder the senescence cycle and related neurodegenerative issues, including cognitive decline and psychological impairment in AD. The mode of action of flavonoids stems from their anti-inflammatory and antioxidative neuroprotective properties.[26,84]

Ellagic acid (EA), a dietary flavonoid, is converted *in vivo* to urolithin metabolites.[85] EA and its urolithin metabolites, UM5 and UM6, exert neuroprotective effects, improving $Aβ_{25–35}$-induced cell damage. Network pharmacology and molecular docking studies demonstrate that EA, UM5, and UM6 play roles in the activation of the PI3K-Akt cell signaling pathway and the inhibition of *mitogen-activated protein kinase*.[85] Specific targeting of Fyn kinase has been proposed as a therapeutic strategy for AD. AZD0530 (Saracatinib), a dual inhibitor of the tyrosine kinases c-Src (proto-oncogene tyrosine-protein kinase) and Abl (Abelson murine leukemia viral oncogene homolog 1), potently inhibits Fyn kinase (non-receptor cytoplasmic proto-oncogene tyrosine-protein kinase), preventing APOE-induced Fyn signaling and downstream phosphorylation of the AD risk gene products *proline-rich tyrosine kinase 2 (*PYK2*)* and tau.[86] PYK2 is a non-receptor tyrosine kinase highly enriched in forebrain neurons and is closely related to focal adhesion kinase, playing important roles in sensing cell-ECM contacts and in ECM signaling that controls cellular adhesion and cell survival. Targeting Fyn kinase has been shown to reverse memory deficits in AD mouse models and to rescue the reduced synaptic density that occurs in AD.[86]

Atomic force microscopy and fluorescent imaging show that 6*O*phosphorylated HS/heparin oligosaccharides inhibit amyloid fibril formation *in vitro* and hold promising *in vivo* applications.[87] The pathogenesis of AD in humans is associated with impaired clearance of extracellular amyloid deposits and the accumulation of insoluble plaque formations. Removal of neuronal HS by conditional deletion of *EXT1* in postnatal neurons leads to a reduction in Aβ oligomerization and deposition of amyloid plaques in APP/S1 mice. An elevation in the HSPG content in AD brains highlights their roles in the pathogenesis of AD and identifies them as potential therapeutic targets.[88] While HSPGs are known to promote plaque formation in AD, HS analogs have been shown to inhibit βsecretase, a transmembrane neuronal aspartic acid protease involved in myelination.[89] The use of enoxaparin (Lovenox) and dalteparin (Fragmin) in a mouse AD model reduced amyloid deposition by re-solubilizing plaques and promoting fibril disassembly. Studies on modified heparins have examined the effects of GAG size and sulfation on interactions with tau, αsyn, and Aβ in terms of cellular uptake and protein aggregation. Tau aggregates have specific requirements for GAG length and sulfation positions, whereas α-syn and Aβ aggregates have less stringent binding requirements.[20]

8.5. Application of HS glycomics in diseases of cognitive decline

HS-like pseudo-hexasaccharides have been prepared by linking HS disaccharides in a "head-to-tail" manner.[90] This new class of compounds binds FGF-2 in a similar manner to native HS oligosaccharides, and they may serve

as useful tools for better understanding HS interactions of physiological significance. Another novel approach in HS glycomimetics involves the use of clickable disaccharide building blocks to assemble oligomers that mimic the disaccharide repeat units of native HS.[91] These HS-mimetic oligomers also bind FGF-2 in a sulfation-dependent manner. To further explore HS structure-function relationships and harness the therapeutic potential of HS, a new class of four trisaccharide compounds derived from the aminoglycoside tobramycin has been identified. These compounds mimic natural HS,[92] exhibiting strong binding to heparanase while displaying low affinity to off-target platelet factor-4, thereby demonstrating their potential as glycopolymers with specific modes of action. Some functions of HS chains, such as receptor activation, require short sulfated sequences separated by *N*-acetylated regions. Advances in HS sequencing, chemical synthesis, and the elucidation of 3D oligosaccharide-protein complex structures have significantly contributed to our understanding of HS structure-function interrelationships. These advancements have led to the development of HS-inspired drugs with therapeutic applications in inflammation and cancer.[93]

HS-mimetic glycopolymers hold promise for regulating protein aggregation processes and may prevent the development of neurodegenerative features in diseases such as AD and PD.

9. Conclusion

This review has highlighted the considerable complexity of HS in both normal and pathological tissues, which endows HSPGs with complex regulatory roles in pathophysiology as well as essential instructive regulatory properties in normal cellular processes. In pathological conditions, such as cancer and neurodegenerative diseases, HSPGs contribute to the propagation of abnormal protein aggregates in neural tissues, leading to disrupted neuronal function. These disruptions have long-term neurotoxic consequences, including neuronal death, cognitive decline, permanent memory loss, motor impairment, and tissue dysfunction, as observed in AD and PD.[94] Factors such as sulfation position, charge density, chain flexibility, and length, and the planar 3D geometric presentation of sulfate groups all contribute to HS's interaction and its role in these processes. Advanced techniques such as X-ray crystallography, in silico molecular docking, quantum semiconductor bio-affinity nanoprobes, sub-nanometer ultra-high resolution live cell imaging, single-cell transcriptomics, neutron scattering, and electron detachment dissociation provide a sophisticated level of analysis. These techniques enhance our predictive capabilities in understanding variations in HS conformation and their effects on HS-ligand recognition and interaction. In addition, artificial intelligence, deep machine learning, and neural network analysis have been applied in the development of interactive algorithms for understanding HS structure and function, and these technologies are being used in the development of HS mimetics with potential therapeutic applications. A greater understanding of HS-protein interactions through these sophisticated techniques will undoubtedly further our understanding of the complexities of this fascinating molecule.[95] However, finding a means to control HS in pathological disorders without disrupting its roles in essential life processes will be a challenging task. If successful, it could have profound benefits for health and well-being.[96] Achieving this goal will require the utilization of next-generation, state-of-the-art technologies, but promising bioactive molecules have already emerged. Ongoing work will hopefully continue to refine these developments, perfecting their application in the challenging fields of pathobiology and bioregulation.

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Conflict of interest

The authors declare no conflicts of interest.

Author contributions

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Writing – original draft: James Melrose
Writing – review & editing: All authors

Ethics approval and consent to participate

Not applicable.

Consent for publication

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Availability of data

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